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Problem Statement

- Chronic stress significantly affects mental health, productivity, and overall well-being worldwide [1].
- Many meditation apps provide generic relaxation experiences without adapting to the user's physiological stress state.

- Wearable devices can monitor stress using signals such as Heart Rate Variability (HRV) but mostly focus on monitoring rather than intervention [2].Existing stress detection models often use binary classification and ignore circadian biological rhythms, leading to inaccurate stress interpretation [3][4].

- Therefore, there is a need for a real-time, physiologically adaptive system that integrates stress detection with personalized meditation interventions.

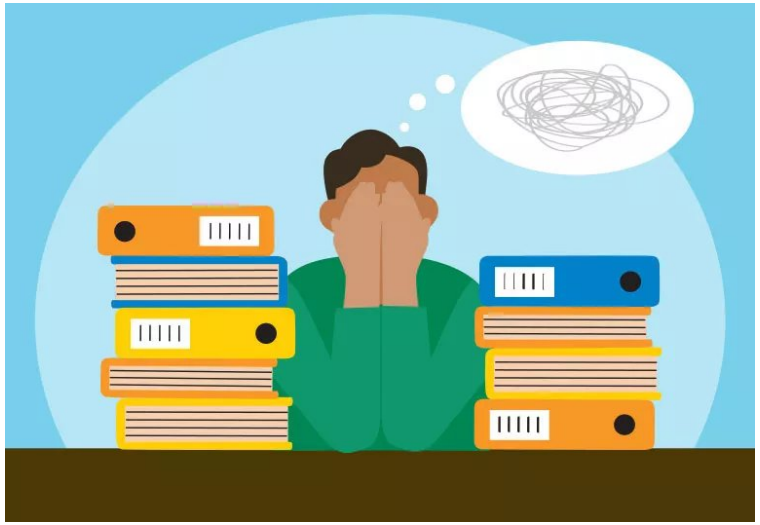
Proposed Solution

- AI – Based Adaptive VR Meditation System

Who Faces This Problem?

Primary Users

- Individuals experiencing daily stress
- Students and professionals with high workload
- People seeking mental wellness support



Key Stakeholders

- Healthcare professionals and therapists
- Mental health researchers
- VR application developers
- Wellness technology companies



Impact of the problem

Impact on users

- Reduced mental wellbeing and concentration
- Increased anxiety and burnout
- Poor sleep and decrease productivity

Impact on Society

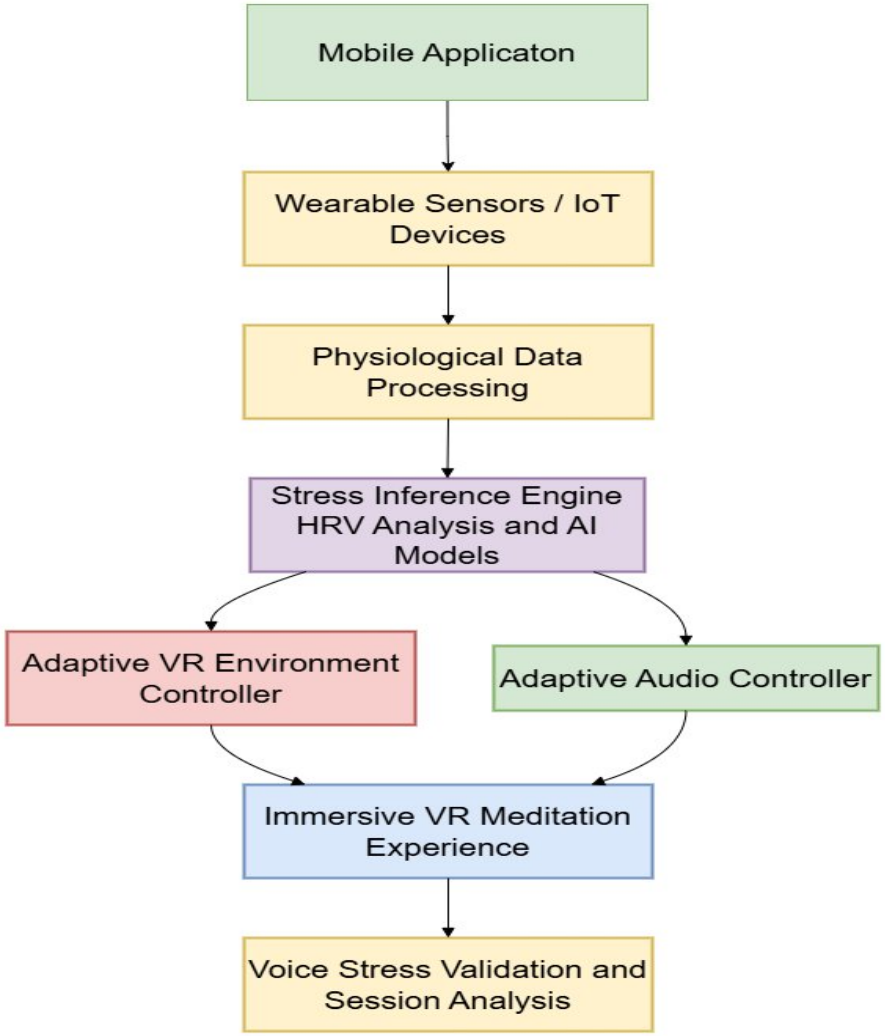
- Increased healthcare costs
- Reduced workplace productivity
- Growing demand for accessible mental health support

Existing Solutions and Their Limitations

Existing Solution	What it does well	Limitation for our context
Meditation Apps (e.g., Calm, Headspace)	Provide guided meditation and relaxation sessions	Sessions are static and not personalized based on the user's real-time physiological stress level
Wearable Stress Monitoring Devices	Monitor physiological signals such as heart rate and HRV	Mainly track stress but do not adapt therapeutic interventions such as meditation environments
HRV-Based Stress Detection Systems	Detect stress from physiological signals using machine learning	Often ignore circadian biological rhythms and rely on binary stress detection
VR Relaxation Environments	Provide immersive environments for relaxation and mindfulness	VR scenes are static and not dynamically adapted to the user's physiological state

GAP: CURRENT STRESS MONITORING AND MEDITATION SYSTEMS OPERATE SEPARATELY AND LACK A CIRCADIAN-AWARE, REAL-TIME ADAPTIVE MECHANISM THAT INTEGRATES PHYSIOLOGICAL STRESS DETECTION WITH PERSONALIZED THERAPEUTIC INTERVENTIONS

Proposed Solution



- **Physiological Data Acquisition** – collects physiological signals from wearable sensors such as HRV and heart rate.
- **Stress Inference Engine** – analyzes HRV features and temporal patterns to classify the user's stress level.
- **Adaptive Environment Controller** – dynamically adjusts meditation visuals and ambience based on detected stress.
- **Adaptive Audio Controller** – modifies soundscape and meditation audio to enhance relaxation.
- **Session Validation Module** – evaluates meditation effectiveness using voice-based stress analysis and system feedback.

CONFIDENCE-AWARE MULTIMODAL RR SENSING VIA PPG QUALITY MODELING & SELECTIVE RECONSTRUCTION

Contribution to the Overall Solution

- Serves as the physiological sensing and signal reliability layer of the overall system
- Acquires wrist-based PPG, accelerometer, EDA, and skin-temperature signals from low-cost wearable hardware
- Detects whether each PPG segment is Reliable, Recoverable, or Unreliable
- Reconstructs only recoverable noisy segments and extracts cleaned RR interval streams
- Transmits confidence-scored cleaned RR intervals to Component B for HRV extraction and stress inference

Novelty

- Introduces a 3-class PPG quality framework instead of conventional binary signal-quality assessment
- Integrates EDA and skin temperature as contextual indicators for PPG quality estimation
- Applies selective reconstruction only to recoverable segments rather than correcting all noisy data
- Outputs a confidence-aware RR stream for robust downstream physiological intelligence

CONFIDENCE-AWARE MULTIMODAL RR SENSING VIA PPG QUALITY MODELING & SELECTIVE RECONSTRUCTION

Problem Definition

- Wrist PPG signals are highly affected by motion artifacts, contact variation, and physiological noise, reducing RR reliability
- Existing signal-quality assessment approaches are mostly binary and do not distinguish recoverable corruption from unusable segments
- Most low-cost sensing pipelines do not integrate multimodal context such as EDA and skin temperature for signal validation
- Noisy RR extraction propagates errors into downstream HRV analysis and stress inference modules such as Component B

Research Objective

- Develop a **confidence-aware multimodal sensing framework** that uses PPG, accelerometer, EDA, and skin temperature to classify signal quality, selectively reconstruct recoverable segments, and generate reliable RR interval streams for downstream stress inference.
- SO1: Collect synchronized wrist-sensor data under rest, motion, and stress conditions
- SO2: Build a 3-class PPG quality classifier for Reliable / Recoverable / Unreliable labeling
- SO3: Develop a 1D U-Net autoencoder for selective reconstruction of recoverable segments
- SO4: Extract and validate cleaned RR interval streams against reference-quality signals
- SO5: Output confidence scores and cleaned RR streams to support Component B

CONFIDENCE-AWARE MULTIMODAL RR SENSING VIA PPG QUALITY MODELING & SELECTIVE RECONSTRUCTION

METHODOLOGY AND TECHNOLOGY STACK

Application of Data Science Domain Knowledge

- **Biomedical Signal Processing** → bandpass filtering, peak detection, RR interval extraction
- **Multimodal Feature Engineering** → PPG waveform features + motion + EDA + skin-temperature context
- **Machine Learning Classification** → 3-class signal quality prediction
- **Deep Learning Reconstruction** → selective recovery of corrupted-but-recoverable PPG segments
- **Confidence Modeling** → confidence-aware output generation for downstream analytics

Methodology

- Acquire multimodal physiological data from wearable sensors
- Segment PPG into short epochs for quality evaluation
- Extract waveform, spectral, motion, EDA, and temperature features
- Classify each segment into Reliable / Recoverable / Unreliable
- Reconstruct only recoverable segments using 1D U-Net
- Perform peak detection and generate cleaned RR interval streams
- Output RR streams + confidence score to Component B

Selected Technologies

- Python as the core implementation platform
- Scikit-learn / RF / SVM for 3-class signal-quality classification
- PyTorch / TensorFlow for 1D U-Net reconstruction model
- NeuroKit2 / custom signal-processing pipeline for peak detection and RR extraction
- Arduino BLE / wearable IoT sensors for physiological data acquisition

CONFIDENCE-AWARE MULTIMODAL RR SENSING VIA PPG QUALITY MODELING & SELECTIVE RECONSTRUCTION

EVALUATION, DATA, SECURITY, AND ETHICS

Data Availability

- Public Benchmark Datasets
- **WESAD**: wrist BVP/PPG, EDA, skin temperature, ACC + stress labels
- **PPG-DaLiA**: wrist PPG, ACC + ECG ground truth
- **IEEE SPC 2015 / IEEEPPG**: 2-channel wrist PPG, ACC, ECG under intense motion
- **PhysioNet Wearable Stress/Exercise Dataset**: BVP, EDA, skin temperature, activity from Empatica E4

Evaluation & Validation

Metrics

- 3-class classification accuracy
- Macro F1-score
- Reconstruction error (RMSE / MAE / signal similarity)
- RR interval accuracy against reference signals

Validation Strategy

- Compare classifier performance across Reliable / Recoverable / Unreliable classes
- Evaluate reconstruction only on recoverable segments
- Assess whether cleaned RR streams improve downstream HRV readiness for Component B

Performance Target

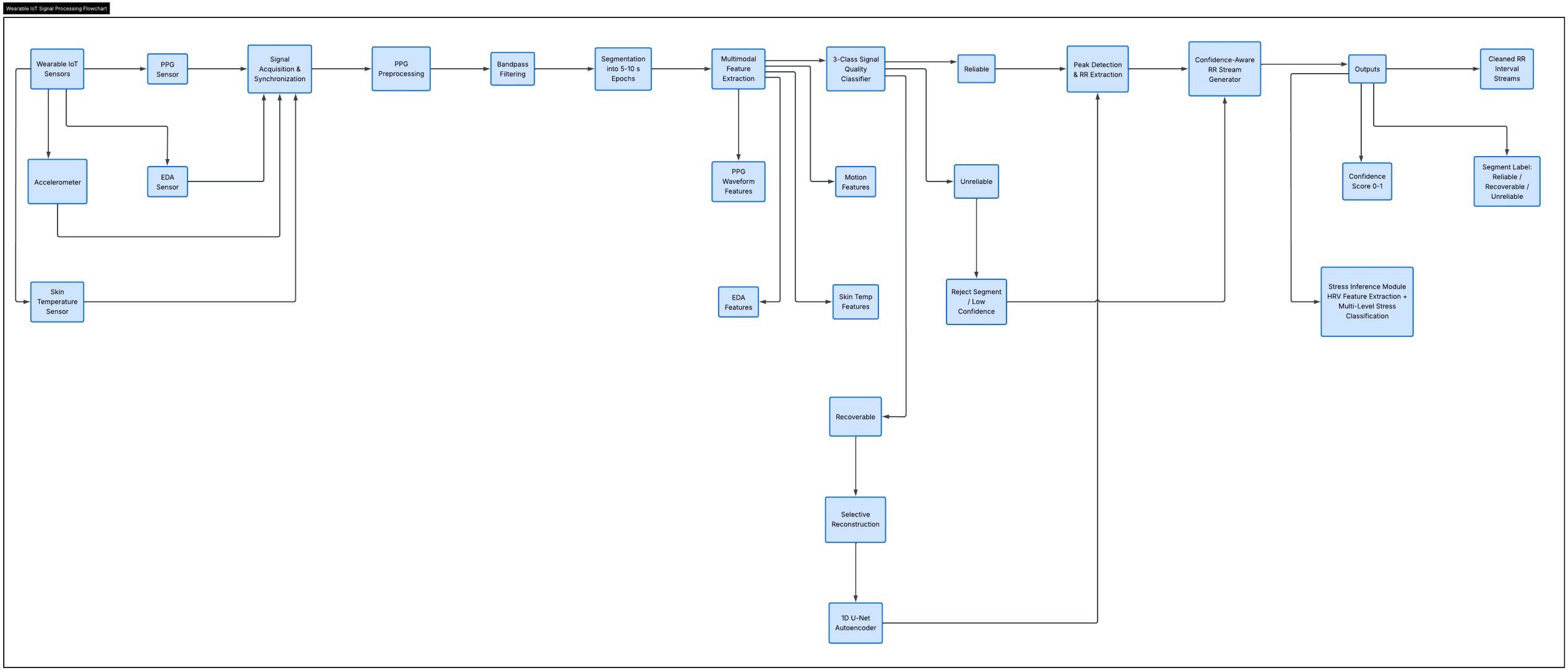
- Achieve strong 3-class signal-quality classification performance
- Improve RR extraction reliability under motion/noise conditions
- Reduce invalid RR propagation into downstream stress inference
- Support near real-time wearable sensing for adaptive VR integration

Security & Ethics

- Achieve strong 3-class signal-quality classification performance
- Improve RR extraction reliability under motion/noise conditions
- Reduce invalid RR propagation into downstream stress inference
- Support near real-time wearable sensing for adaptive VR integration

CONFIDENCE-AWARE MULTIMODAL RR SENSING VIA PPG QUALITY MODELING & SELECTIVE RECONSTRUCTION

SYSTEM OVERVIEW AND INTEGRATION



Dependencies

- Wearable IoT sensing hardware for PPG, accelerometer, EDA, and skin temperature
- **Component B**: Multi-level stress inference via HRV modeling
- **Component C** / overall VR adaptation pipeline for final system response

Output

- Cleaned RR interval streams
- Segment-wise confidence scores
- Signal-quality labels: Reliable / Recoverable / Unreliable
- Real-time physiological signal reliability output sent to Component B

Role in AI-Based VR Meditation System:

This component acts as the **signal quality assurance and RR generation layer** of the system, ensuring that only confidence-scored, cleaned RR interval streams are passed to **Component B** for HRV feature extraction and multi-level stress inference.

IT22191038 | PERERA B.D.P.D.

MULTI – LEVEL STRESS INFERENCE VIA TEMPORAL PATTERN MODELING



Contribution to the Overall Solution

- Serves as the AI stress inference engine (Physiological Intelligence Layer) of the system
- Receives cleaned RR interval streams from Component A's wearable IoT sensing pipeline
- Extracts time-domain and frequency-domain HRV features (RMSSD, SDNN, LF/HF ratio)
- Classifies physiological states into four stress levels: Relaxed, Mild, Moderate, High
- Transmits real-time stress predictions Component C's adaptive VR environment

Novelty

- Integrates circadian rhythm modeling with machine learning
- Uses Cosinor-Based Rhythm Modelingto model 24-hour HRV patterns
- Removes circadian physiological noise before stress classification

MULTI – LEVEL STRESS INFERENCE VIA TEMPORAL PATTERN MODELING

Problem Definition

- HRV varies naturally across circadian biological cycles
- Existing systems apply static population-based thresholds, causing false stress detection during normal physiological dips
- Binary stress/no-stress classification is insufficient for nuanced therapeutic VR interventions requiring differentiation across multiple stress levels
- Current models exhibit "brittleness" which means, trained on controlled datasets (e.g., WESAD), they fail to generalize across different sensors, datasets, and user

Research Objective

- Develop a temporally-aware HRV stress inference engine that integrates circadian rhythm modeling and machine learning to accurately classify stress levels.

Sub Objectives

- Model circadian HRV baseline using Cosinor rhythmometry
- Extract circadian residual features
- Develop CNN-LSTM model for multi-level stress classification
- Enable real-time stress inference from wearable data
- Validate model using physiological datasets

MULTI – LEVEL STRESS INFERENCE VIA TEMPORAL PATTERN MODELING

METHODOLOGY AND TECHNOLOGY STACK

Application of Data Science Domain Knowledge

- **Feature Engineering** → HRV time-domain and frequency-domain feature extraction
- **Temporal Data Modeling** → capturing time-based physiological patterns
- **Mathematical Modeling** → Cosinor rhythmometry for circadian trend estimation
- **Machine Learning Classification** → multi-class stress prediction
- **Model Generalization Techniques** → cross-dataset validation and robustness analysis

Methodology

- HRV feature extraction from RR interval streams
- Temporal context integration using time-of-day data
- Circadian rhythm modeling using Cosinor analysis
- Residual feature generation (detrended HRV signals)
- Machine learning based multi-level stress classification

Selected Technologies

- NeuroKit2 for HRV signal analysis
- Cosinor modeling for circadian rhythm estimation
- XGBoost and CNN-LSTM deep learning models
- Python ML ecosystem (PyTorch, Scikit-learn)
- Python- Core platform for implementation

MULTI – LEVEL STRESS INFERENCE VIA TEMPORAL PATTERN MODELING

EVALUATION, DATA, SECURITY, AND ETHICS

Data Availability

- Public Benchmark Datasets
- WESAD dataset for physiological stress detection
- SWELL-KW dataset for stress in knowledge work environments
- TILES-2018 dataset for longitudinal physiological monitoring
- IoT Physiological Data collected from wearable sensors in Component A.

Evaluation & Validation

Metrics

- Accuracy
- Macro F1-score
- Confusion Matrix

Validation Strategy

- Cross-dataset validation to test model generalization
- Model comparison between XGBoost, LSTM, and CNN-LSTM models

Performance Target

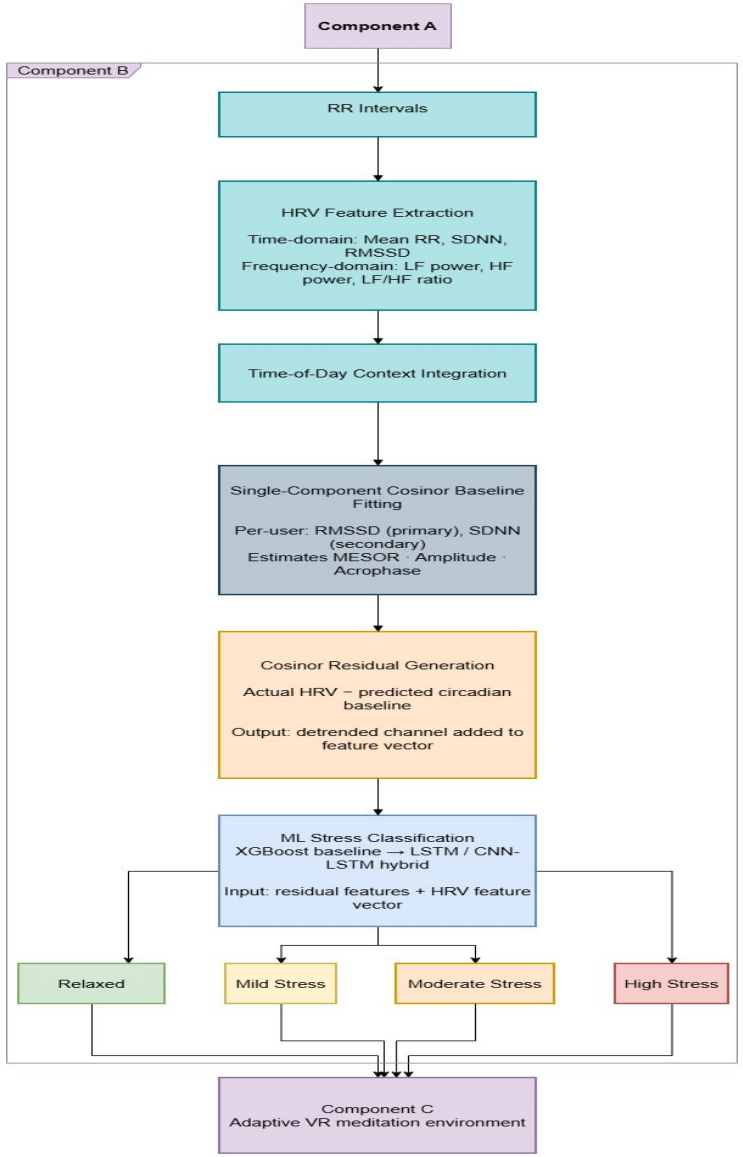
- Achieve ≥ 85% macro F1-score for four-level stress classification
- Maintain high robustness across datasets (WESAD, SWELL-KW, TILES-2018)
- Reduce false stress detections caused by circadian HRV variations
- Support near real-time stress inference for adaptive VR responses

Security & Ethics

- Physiological data anonymization
- Secure storage of time-series data
- Ethical approval and informed participant consent

MULTI – LEVEL STRESS INFERENCE VIA TEMPORAL PATTERN MODELING

SYSTEM OVERVIEW AND INTEGRATION



Dependencies

- Component A Wearable IoT physiological sensing
- Component C Adaptive VR meditation environment

Output

- 4-level stress classification (Relaxed, Mild, Moderate, High)
- Stress probability scores
- Real-time prediction from HRV signals
- Sent to VR adaptation module (Component C)

Role in AI-Based VR Meditation System:

This component acts as the AI intelligence layer that analyzes HRV features from wearable sensors to infer the user's stress level and provide real-time stress signals to the VR adaptation system.

IT22231314 | PERERA W.T.S.R.

RL-Driven Personalized VR Environment Parameter Optimization for User Relaxation



Contribution to the Overall Solution

- Serves as the core VR component of the system
- Receives stress classifications from Component B
- Receives user's preferences from the mobile application (Cloud hosted backend)
- Baseline rule-based controller before RL agent takes over
- RL agent learns changing which parameters has the highest impact on user relaxation overtime through state -> action -> reward RL learning process

Novelty

- Introduces a Reinforcement Learning framework that explicitly models pluralistic and divergent user relaxation preferences in VR environments

RL-Driven Personalized VR Environment Parameter Optimization for User Relaxation

Problem Definition

- VR environments are increasingly used for stress relief and relaxation
- Most VR relaxation systems present static VR environments or rely on simple biofeedback adjustments
- These systems do not dynamically learn how different environmental parameters affect the relaxation levels of individual users
- Since human responses to relaxation stimuli vary significantly, there is a need for an intelligent system capable of learning and adapting VR environment parameters in real-time to optimize relaxation of individual users

Research Objective

- Develop a Reinforcement Learning-Based system that dynamically optimizes VR environment parameters using physiological feedback to maximize user relaxation while handling pluralistic user preferences

Sub Objectives

- Define controllable VR parameters influencing relaxation
- Define a baseline rule-based controller to initialize the environment before RL agent takes over
- Develop a RL agent to optimize environment settings for optimal user relaxation
- Handle pluralistic and dynamic user relaxation preferences

RL-Driven Personalized VR Environment Parameter Optimization for User Relaxation

METHODOLOGY AND TECHNOLOGY STACK

Application of Data Science Domain Knowledge

- **AI and RL** → RL-based framework for parameter optimization
- **Generative AI** → For generating the base VR environment
- **Prompt Engineering** → Passing user preferences to the GenAI model through well-structured prompts

Methodology

- User preference collection through the mobile application
- Apply prompt engineering to convert collected user preferences into structured prompts
- Use GenAI models to generate the initial VR environment
- Generated VR environment serves as the starting state for the adaptive system
- Stress classification model outputs become a part of the state input for the RL agent
- Implement DRL agent. Define,
  State -> user stress level + current VR parameters
  Actions -> adjust environment parameters
  Reward -> Reduction in stress
- RL agent sends the parameters to Unity engine
- Unity engine adjusts the parameters in real-time
- Data logging, system monitoring and system evaluation

Selected Technologies

- Unity as the VR environment controller engine
- Runway GWM-1 world model for generating the base VR environment
- PyTorch for the neural networks required for RL
- REST API for communication between RL agent and VR engine

RL-Driven Personalized VR Environment Parameter Optimization for User Relaxation

EVALUATION, DATA, SECURITY, AND ETHICS

Data Availability

- User preference data collected through the mobile application
- Interaction data generated during VR sessions
- Stress classification data received from Component B

Evaluation & Validation

User Experience Metrics

- Self-reported relaxation score
- Perceived immersion level

Validation Strategy

- Collect baseline physiological data (Stress level) before the VR session
- Expose users to both static and RL-adaptive environments
- Record post-session data and feedback from participants

Performance Target

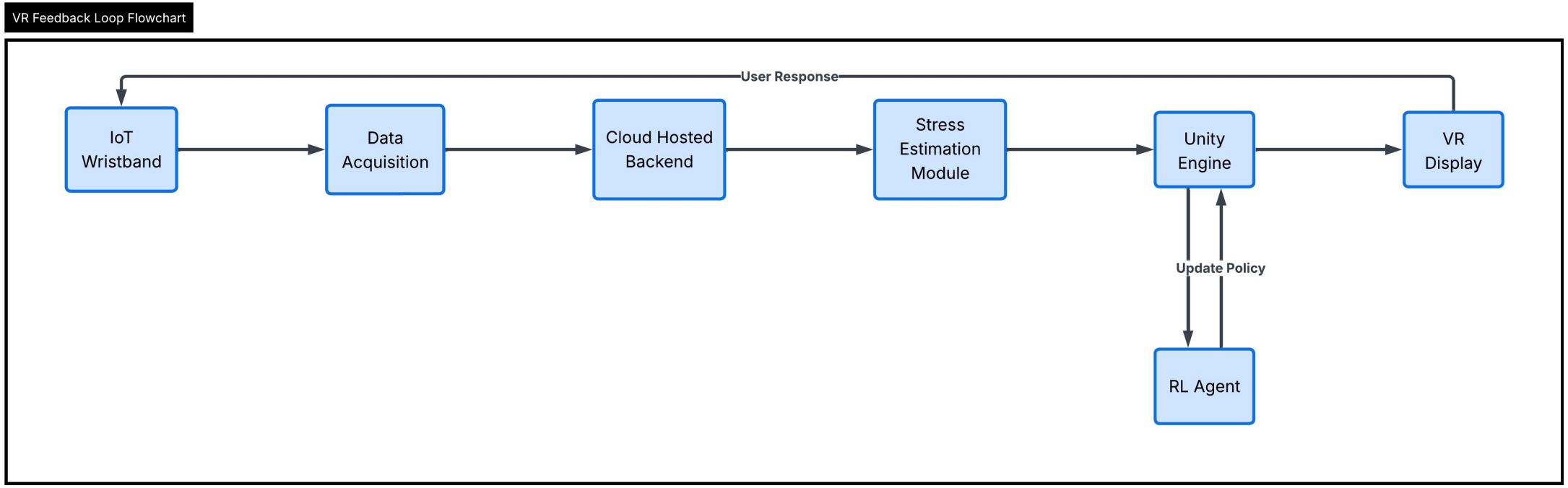
- VR environment adaptation within 1-2 seconds after stress detection
- Measurable reduction in user stress during VR session
- High personalization accuracy
- Stable RL agent learning performance with consistent reward improvement over training sessions

Security & Ethics

- Secure storage of collected physiological and user preference data
- Anonymization of user data to protect participant identity
- Responsible AI: RL system will only adjust environmental relaxation parameters, avoiding manipulative or harmful interventions

RL-Driven Personalized VR Environment Parameter Optimization for User Relaxation

SYSTEM OVERVIEW AND INTEGRATION



Dependencies

- Component B – Stress classification data
- Mobile application – User preferences

Output

- RL-driven adaptive VR environment

Role in AI-Based VR Meditation System

This component acts as the core VR component that facilitates optimal user relaxation through reinforcement learning-driven parameter optimization.

IT22272396 | SAMARASEKARA K. S. M.

Personalized Adaptive Audio Controller



Contribution to the Overall Solution

- Generates personalized meditation sound-wave layers using Google Lyria Realtime
- Controls sound-wave parameters and ambient-audio blending using user stress level and Component A confidence score
- Reduces abrupt or unsafe adaptation when confidence is low
- Produces a confidence-aware adaptive therapeutic audio stream for the VR meditation environment

Novelty

- Personalized AI-generated therapeutic audio for each user
- Confidence-aware, stress-driven real-time adaptation
- Dual-layer audio design: ambience + Lyria personalized layer

Personalized Adaptive Audio Controller

Problem Definition

- Most VR meditation systems use static background music or fixed soundscapes
- Existing systems rarely adapt both generated meditation audio and ambient VR audio in real time
- Audio adaptation should depend not only on stress level, but also on the reliability of physiological input
- Without confidence-aware control, the system may overreact to noisy stress signals and produce unstable or uncomfortable sound changes

Research Objective

- SO1: Design a personalized adaptive audio control framework for AI-generated meditation sound waves
- SO2: Build a user preference profile to guide personalized sound-wave generation and ambient audio behavior
- SO3: Develop a confidence-aware stress-driven control policy to adjust meditation sound parameters and ambient-audio mix in real time
- SO4: Evaluate the system against static and non-personalized audio baselines using relaxation, immersion, stability, and latency metrics

Personalized Adaptive Audio Controller

METHODOLOGY AND TECHNOLOGY STACK

Application of Data Science Domain Knowledge

- **Preference Modeling** → capturing user-specific audio preferences for personalization
- **Reinforcement Learning** → learning user-specific audio adaptation policies over time
- **Confidence-Aware Decision Making** → moderating adaptation using signal reliability
- **Affective Computing** → mapping stress state to calming audio control strategies
- **Adaptive Audio Parameter Control** → adjusting volume, density, brightness, pulse, and fade behavior

Methodology

- Collect user audio preferences through the mobile application
- Build a personalized prompt profile for meditation sound-wave generation
- Generate personalized sound layers using Google Lyria Realtime
- Receive stress level from Component B and confidence score from Component A as runtime control inputs
- Use confidence-aware rule constraints + RL-based adaptation to control audio parameters such as intensity, density, brightness, tempo, fade, and mix level
- Blend the personalized sound-wave layer with VR ambient audio inside Unity/Unreal using Wwise or FMOD

Selected Technologies

- NeuroKit2 for HRV signal analysis Google Lyria RealTime for personalized generative meditation audio
- Unity / Unreal Engine for integration and runtime control
- Wwise or FMOD for adaptive audio mixing and transitions
- Python + RL libraries for training the personalization agent
- Mobile App for preference collection and session feedback

Personalized Adaptive Audio Controller

EVALUATION, DATA, SECURITY, AND ETHICS

Data Availability

- Public Benchmark Datasets
- User preference data collected through the mobile app
- Stress level from Component B and confidence score from Component A
- Session feedback data for personalization and RL policy improvement
- IoT Physiological Data collected from wearable sensors in Component A
- Runtime interaction logs such as audio adjustments, mix levels, and session states.

Evaluation & Validation

Metrics

- Relaxation improvement
- Immersion / user experience score
- Audio adaptation stability
- System latency

Validation Strategy

- Compare against static audio and non-personalized audio baselines
- Evaluate RL-based adaptation vs fixed rule-based control
- Conduct user studies with questionnaire-based feedback and session analysis

Performance Target

- Achieve smooth real-time adaptation without abrupt audio changes
- Maintain low-latency audio response suitable for VR sessions
- Improve relaxation and immersion compared with baseline audio methods
- Ensure stable behavior when confidence is low

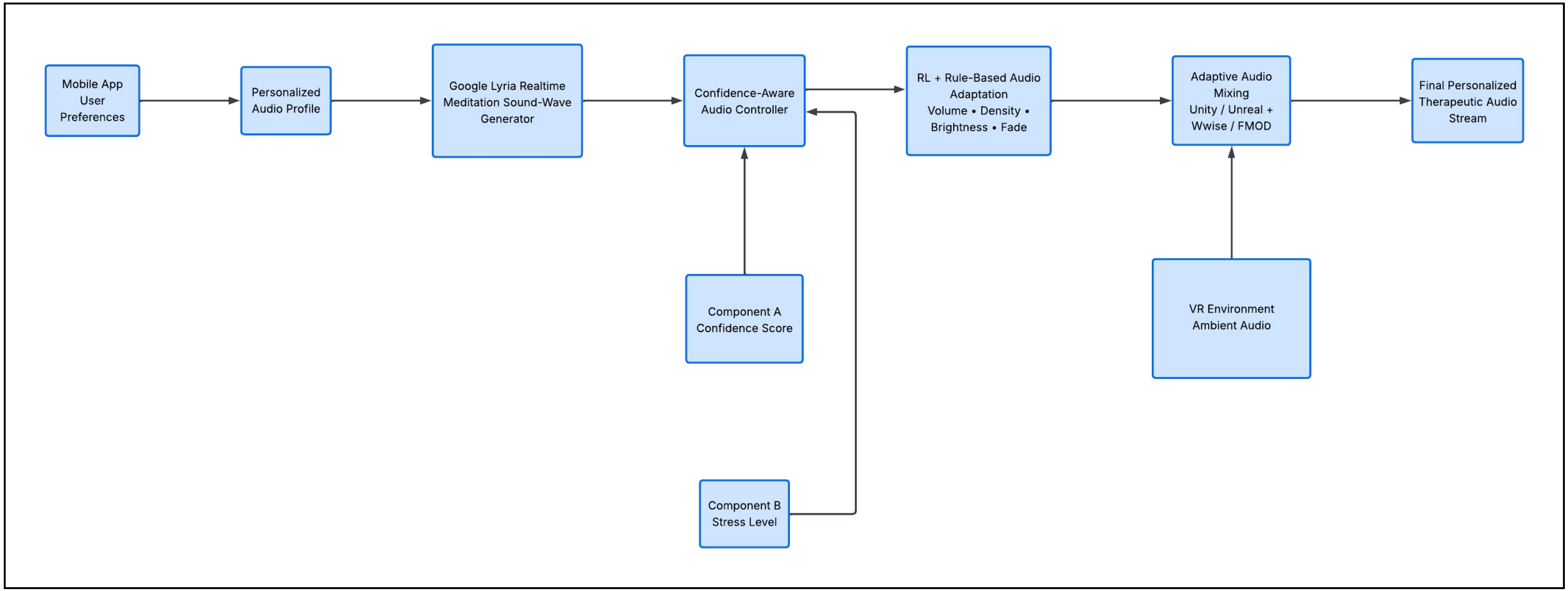
Security & Ethics

- Secure storage of user preferences, stress data, and feedback
- Use anonymization/pseudonymization for participant data
- Obtain ethical approval and informed consent for user testing
- Avoid over-adaptation that may cause discomfort during meditation sessions

Personalized Adaptive Audio Controller

SYSTEM OVERVIEW AND INTEGRATION

Adaptive Audio Flow Diagram



Personalized Adaptive Audio Controller

SYSTEM OVERVIEW AND INTEGRATION

Dependencies

- **Component A**: signal confidence / wearable physiological reliability input
- **Component B**: user stress level input
- **Component C**: VR environment ambient audio layer
- **Mobile App**: user preference collection

Output

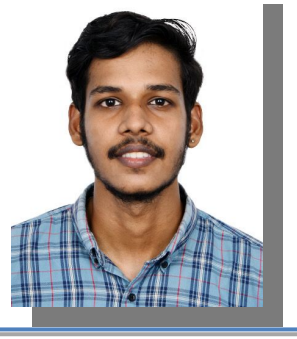
- Personalized meditation sound-wave layer generated through Google Lyria RealTime
- Adaptive audio control decisions based on stress level and confidence score
- Balanced final audio mix with VR ambient audio in Unity/Unreal
- Real-time personalized audio stream sent to the VR system

Role in AI-Based VR Meditation System:

This component acts as the adaptive therapeutic audio controller of the system, generating personalized meditation sound waves and dynamically controlling their behavior and ambient-audio blending using user stress level from Component B and physiological confidence from Component A.

IT22621788 | PRATHIKESH S

Voice-Based Multilingual Stress Assessment and Cross-Modal Session Validation in VR Meditation Systems



Contribution to the Overall Solution

- Serves as the session evaluation layer of the AI-based stress-aware VR meditation system
- Collects pre/post-session voice recordings via mobile application
- Scores acoustic stress using fine-tuned XLSR-53 cross-lingual speech model
- Validates acoustic scores against HRV physiological outputs from Component B through cross-modal validation
- Applies Autoencoder-based anomaly detection to flag ineffective meditation sessions
- Delivers a complete session evaluation report to the mobile application

Novelty

- First voice-based session evaluation mechanism integrated into a consumer-grade VR meditation system
- Cross-modal validation combining acoustic with physiological stress signals — no existing VR system achieves this
- Multilingual: English (current) → Sinhala (in development) → Tamil (future extension)
- Autoencoder anomaly detection trained on normal sessions only — no labelled failure data required.

Voice-Based Multilingual Stress Assessment and Cross-Modal Session Validation in VR Meditation Systems

Problem Definition

- No existing VR meditation system evaluates session effectiveness through voice
- All published voice stress detection systems are English-only — no Sinhala or South Asian language support exists
- No VR wellness platform includes automated session-level anomaly detection for ineffective sessions
- Single-modality systems cannot detect cognitive stress persisting in voice while the body physically relaxes

Research Objective

- Develop a multilingual voice-based acoustic stress assessment module with cross-modal session validation for a consumer-grade VR meditation system — achieving a minimum stress classification accuracy of **75%** across English with Sinhala speakers.

Sub Objectives

- Pre-session ambient environment assessment — RMS energy with Zero Crossing Rate
- Pre/post acoustic stress comparison — detect session-level vocal stress reduction
- Cross-modal validation — acoustic scores vs HRV physiological outputs from Component B
- Autoencoder anomaly detection — flag sessions deviating from normal stress reduction patterns

VOICE-BASED MULTILINGUAL STRESS ASSESSMENT WITH CROSS-MODAL SESSION VALIDATION

METHODOLOGY AND TECHNOLOGY STACK

Application of Data Science Domain Knowledge

- **Signal Processing** → RMS energy with Zero Crossing Rate for environment assessment
- **Cross-Lingual Speech AI** → XLSR-53 fine-tuned for acoustic stress classification
- **Affective Computing** → pre/post acoustic score comparison for stress change detection
- **Multimodal Fusion** → acoustic with HRV cross-modal validation framework
- **Unsupervised ML** → Autoencoder anomaly detection via reconstruction error

Methodology

- Ambient noise assessment using RMS with ZCR
- Voice recording — 16 kHz resampling, silence trimming, normalisation
- XLSR-53 fine-tuning — binary stress classification
- Pre/post acoustic score comparison — stress change detection
- Cross-modal validation — acoustic scores vs HRV from Component B
- Autoencoder — MSE reconstruction error flags anomalous sessions

Selected Technologies

- XLSR-53 (Wav2Vec 2.0) — cross- lingual speech foundation modal.
- PyTorch — model training with fine-tuning framework
- Librosa / Torchaudio — audio preprocessing
- Python — core implementation preprocessing

VOICE-BASED MULTILINGUAL STRESS ASSESSMENT WITH CROSS-MODAL SESSION VALIDATION

EVALUATION, DATA, SECURITY, AND ETHICS

Data Availability

- Public Benchmark Datasets
- RAVDESS — English speech emotion corpus (1,440 recordings)
- CREMA-D — English speech emotion corpus (91 actors · 7,442 recordings)
- Locally Collected- Sinhala stress voice dataset — 30–40 university student volunteers

Evaluation & Validation

Metrics

- Accuracy
- Precision · Recall · F1-score (per language)
- AUC-ROC (Autoencoder anomaly detection)

Validation Strategy

- 80-10-10 train/validation/test split — stratified by language with stress label
- 5-fold cross-validation during fine-tuning

Performance Target

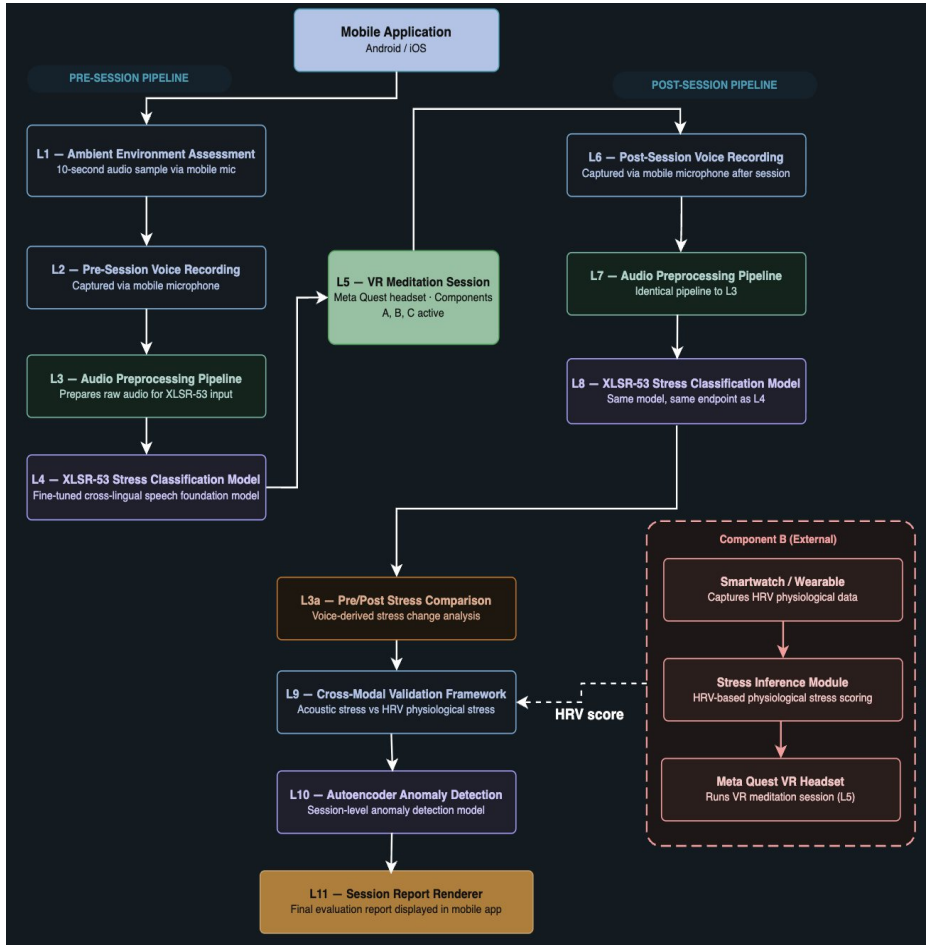
- Achieve ≥ 75% stress classification accuracy across English with Sinhala speakers
- Autoencoder AUC-ROC ≥ 0.75 on session anomaly evaluation dataset
- Cross-modal validation agreement rate reported across all evaluation sessions
- Compare all results against baseline threshold-based approach

Security & Ethics

- Voice recordings anonymized immediately upon receipt
- Raw audio auto-deleted from AWS S3 after stress score generation
- Ethical approval and informed participant consent

VOICE-BASED MULTILINGUAL STRESS ASSESSMENT WITH CROSS-MODAL SESSION VALIDATION

SYSTEM OVERVIEW AND INTEGRATION



Dependencies

- Component B — HRV physiological stress scoring (cross-modal validation input)
- Component C — Adaptive VR meditation environment (receives session evaluation output)

Output

- Pre/post acoustic stress scores (0–10 scale)
- Cross-modal agreement status — Agreement / Mismatch flag
- Session anomaly status — Normal / Anomalous
- Complete session evaluation report delivered to mobile application

Role in AI-Based VR Meditation System:

This component acts as the **session evaluation intelligence layer** — scoring acoustic stress through fine-tuned XLSR-53, validating against HRV signals from Component B, detecting ineffective sessions to ensure measurable, evidence-based stress relief outcomes.

AI Based Adaptive VR Meditation System

Project / Product Name	**AI-Based Adaptive VR Meditation System**
Commercial Product Concept	A personalized digital wellness platform that combines wearable physiological sensing, AI-driven stress inference, adaptive VR environment control, personalized therapeutic audio, and session validation to deliver more effective meditation and stress-reduction experiences.
Core Market Problem	Existing meditation apps and VR relaxation systems are mostly static, generic, and rarely adapt to the user's real stress state, signal confidence, or changing personalization needs.
Overall Value Proposition	Provides a closed-loop therapeutic meditation experience that measures the user's physiology, infers stress in real time, adapts VR and audio intelligently, and validates session effectiveness afterward. This creates a more personalized, measurable meditation product than standard wellness apps.

AI Based Adaptive VR Meditation System

Primary Target Customers	1. Individual consumers seeking stress relief and guided meditation 2. Corporate wellness programs 3. Mental health / digital therapeutics providers 4. Hospitals, therapy centers, and rehabilitation clinics 5. Universities and research labs studying stress, VR, and biofeedback
Secondary Target Customers	AI wellness studios, meditation coaches, health-tech startups, insurance-backed preventive wellness programs, and wearable-device ecosystem partners
Customer Benefits	Higher personalization, better stress-reduction potential, real-time adaptation, safer control through confidence awareness, measurable session outcomes, multilingual accessibility, and stronger trust than static meditation systems.

AI Based Adaptive VR Meditation System

Delivery Model	Mobile app + VR headset experience + wearable sensor integration + cloud analytics/dashboard for personalization and reporting
Business Model	B2C subscription for end users, B2B licensing for clinics and enterprises, and SDK/API licensing for third-party wellness or VR development
Revenue Streams	1. Monthly / annual premium subscription 2. Enterprise wellness licensing 3. Clinical platform licensing 4. White-label SDK/API licensing 5. Hardware bundle revenue with compatible wearable partners 6. Research / institutional deployment packages
Pricing Approach	Consumer tier: low monthly subscription. Clinical / enterprise tier: higher-seat or annual license pricing. SDK/API tier: usage-based or developer licensing model.
Scalability	The product can scale through cloud-hosted analytics, modular APIs, cross-platform mobile integration, and support for multiple industries such as wellness, education, therapy, and digital health.

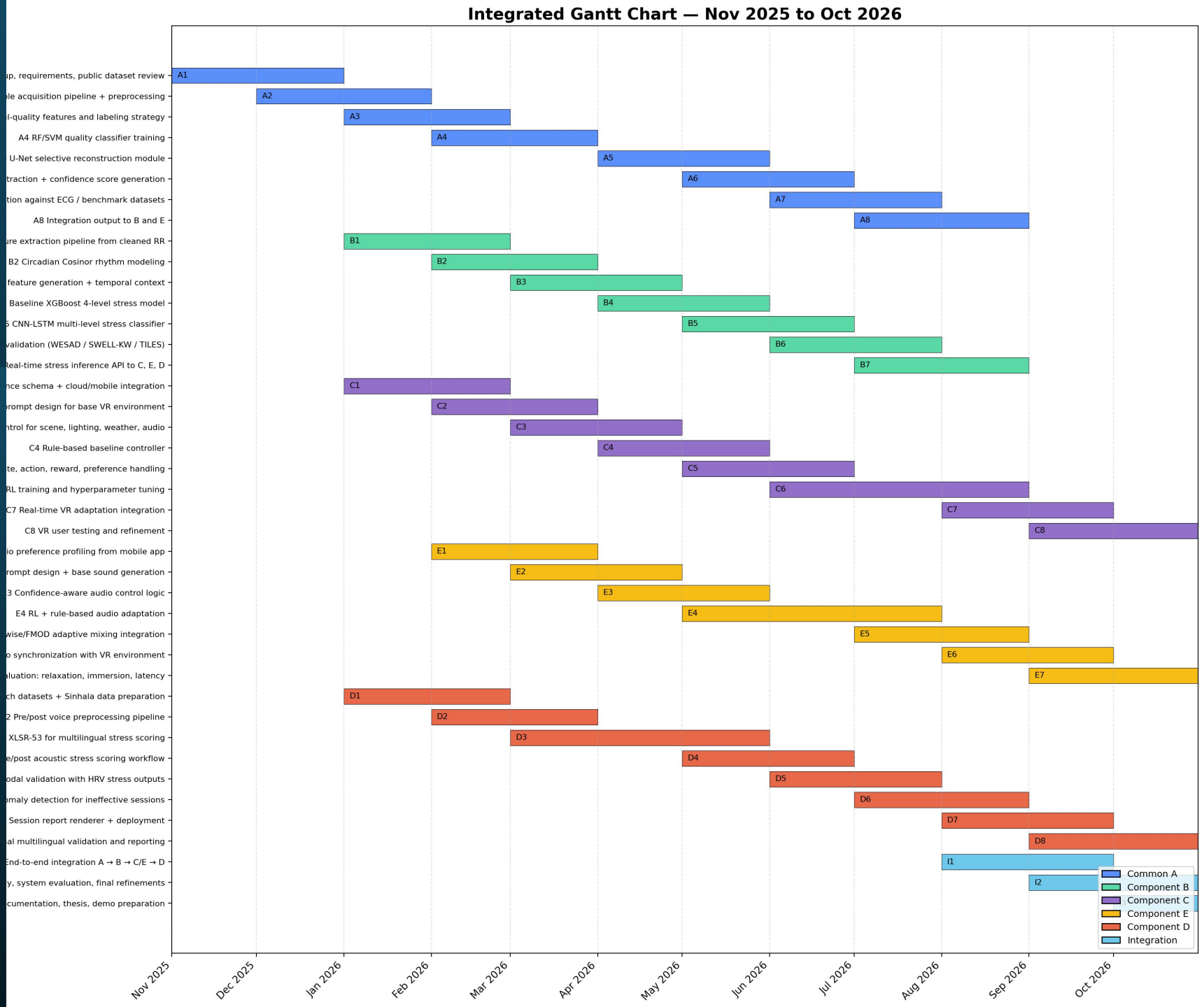
AI Based Adaptive VR Meditation System

Competitive Advantage	Unlike normal meditation apps, this system combines confidence-aware sensing, multi-level stress intelligence, RL-driven adaptive VR personalization, multilingual voice-based session validation, and adaptive therapeutic audio control into one integrated product.
Go-to-Market Strategy	Start with a pilot MVP for universities, wellness centers, and small therapy clinics. Then expand into corporate wellness and consumer subscriptions once the adaptive models and usability are validated.
Pilot Deployment Strategy	Conduct pilot trials with student populations, office workers, and clinic users. Use these pilots to validate user relaxation, immersion, adaptation stability, and session outcome reporting.
Commercial Feasibility	The project is commercially feasible because it is built on a combination of software intelligence + affordable hardware + subscription-ready wellness delivery, which is scalable beyond a research prototype.
Key Partners	Wearable device manufacturers, VR hardware vendors, Unity/Unreal developers, wellness app companies, clinics, universities, corporate HR/wellness providers, and cloud/AI infrastructure partners.

AI Based Adaptive VR Meditation System

Component	Commercial Role
Component A	**Confidence-Aware Sensing Layer** — Can be commercialized as a wearable signal-quality engine / SDK for low-cost wrist-based physiological sensing. It improves reliability by sending confidence-aware cleaned signals to downstream models, making the full system more usable with affordable hardware.
Component B	**Stress Intelligence Engine** — Can be commercialized as a real-time physiological stress inference API for wellness apps, VR systems, and biofeedback tools. It converts cleaned RR signals into four-level stress classification and probability scores.
Component C	**Adaptive VR Personalization Engine** — This is the main interactive product layer and can be commercialized as a VR adaptation module / SDK for relaxation and mental wellness applications. It personalizes environment parameters according to stress level and user preference.
Component D	**Session Validation and Outcome Analytics Layer** — Can be commercialized as a post-session effectiveness analytics module, especially for clinics, enterprises, and researchers who need measurable evidence of therapeutic benefit. Its multilingual voice-based validation and anomaly detection increase trust and accountability.
Component E	**Personalized Adaptive Therapeutic Audio Engine** — Can be commercialized as a generative meditation audio controller / API that adapts sound using stress level and physiological confidence. It is valuable as a standalone audio personalization product or as part of the full VR meditation platform.

AI Based Adaptive VR Meditation System



AI Based Adaptive VR Meditation System

- THANK YOU